

Swarm Density Measures and the Connectivity Threshold

Reference notes for the Unity swarm perception testbed — how the density end condition is computed, and how to choose its threshold.

1. What is being measured

A recording can end when the swarm has changed shape by a set amount. Two measures of “spread” are available, both implemented in `SwarmDensityMetrics.cs`:

Measure	Symbol	Units	What it is
Trimmed hull area	A	u^2	Area of the convex hull of the agents, after discarding the most peripheral fraction (default 10%) so a single straggler cannot dominate it.
Mean KNN distance	d	u	Each agent's mean distance to its K nearest neighbours ($K = 3$), averaged over the swarm. This is the intra-group distance measure used by Hénard et al. (2024), section 3.2.1.

Both shrink as the swarm contracts. The end condition compares the current value against a baseline captured at the start of the recording, so the trigger is a dimensionless ratio:

$\text{ratio} = \text{current} / \text{baseline}$

$\text{ratio} < 1$ the swarm has contracted

$\text{ratio} > 1$ the swarm has expanded

Area and distance ratios are not interchangeable

Area is length squared, so a contraction that halves every distance quarters the area:

$\text{area ratio} = (\text{distance ratio})^2$

A hull target of 0.40 is the same physical contraction as a KNN target of $\sqrt{0.40} = 0.63$. Setting 0.40 on the KNN metric asks for a far tighter clump.

2. Density

Density is agents per unit of floor area:

$$\lambda = N / A$$

For the Densification spawn area in `Scene6_Expand` ($18.17 \times 8.13 = 147.7 \text{ u}^2$) with $N = 40$ agents:

$$\lambda = 40 / 147.7 = 0.271 \text{ agents per } u^2$$

3. Mean degree: how many neighbours an agent perceives

An agent perceives everything within its perception radius R , which is a disc of area πR^2 . Multiplying that area by the density gives the expected number of agents caught inside it. In graph terms each perceived neighbour is an edge, so this count is the *degree* of the node:

$$\text{mean degree } k = \lambda \times \pi R^2 = (N / A) \times \pi R^2$$

With $R = 1.7$ (the Flocking and Densification preset):

$$\text{disc area} = \pi \times 1.7^2 = 9.08 \text{ u}^2$$

$$\text{mean degree} = 0.271 \times 9.08 = 2.46 \text{ neighbours}$$

4. The connectivity threshold

Draw an edge between every pair of agents that can perceive each other and you have the *perception graph*. Hénard et al. define swarm fragmentation by this graph breaking into disconnected components. As density rises the graph passes through a sharp transition, from many small components to a single spanning one, at a mean degree of approximately **4.51**. Section 8 gives the provenance of that number.

Mean degree	Regime
about 1	Agents pair off. Many isolated islands.
about 3	Short chains, still broken into separate clumps.
about 4.51	Percolation transition: clumps link into one spanning structure.
about 8	Densely connected throughout.

Setting the mean degree equal to the threshold and solving for area gives the *critical area*, the footprint at which the swarm sits on the edge of cohesion:

$$\begin{aligned}\lambda \times \pi R^2 &= 4.51 \\ (N / A) \times \pi R^2 &= 4.51 \\ N \times \pi R^2 &= 4.51 \times A\end{aligned}$$

$$A_c = N \times \pi R^2 / 4.51$$

For $N = 40$ and $R = 1.7$:

$$A_c = 40 \times 9.08 / 4.51 = 363.2 / 4.51 = 80.5 \text{ u}^2$$

Spread wider than 80.5 u^2 and the swarm tends to fragment; tighter and it holds together as one.

5. Converting between the two measures

In a random scatter, typical spacing scales as $1/\sqrt{\lambda}$, that is $\sqrt{(A/N)}$: double the density and distances shrink by $\sqrt{2}$. The constant in front comes from the standard expected distances to successive nearest neighbours.

Neighbour	Mean distance
1st nearest	$0.500 \times \sqrt{(A/N)}$
2nd nearest	$0.750 \times \sqrt{(A/N)}$
3rd nearest	$0.938 \times \sqrt{(A/N)}$
average of the three (K = 3)	$0.729 \times \sqrt{(A/N)}$

$$\begin{aligned}d &\approx 0.729 \times \sqrt{(A / N)} \text{ (distance from area)} \\ A &\approx N \times (d / 0.729)^2 \text{ (area from distance)}\end{aligned}$$

Check against the Densification spawn area: $0.729 \times \sqrt{(147.7/40)} = 0.729 \times 1.922 = 1.40 \text{ u}$, matching the measured value in the table below.

6. The Scene6_Expand numbers

Arena: x from -20.20 to 22.23 , y from -12.60 to 11.18 , so $42.43 \times 23.78 \approx 1009 \text{ u}^2$. All rows use $N = 40$. A_c/A is how far the swarm must change, as an area ratio, to reach the connectivity threshold; the KNN column is the same target expressed for the distance metric, that is $\sqrt{(A_c/A)}$.

Behaviour	Spawn area	R	Mean degree	Mean 3NN	A_c	A_c / A	as KNN
Densification	147.7 u ² (18.2×8.1)	1.7	2.46	1.40	80.5	0.55	0.74
Flocking	59.1 u ² (3.4×17.3)	1.7	6.15	0.89	80.5	1.36	1.17
Random	197.7 u ² (20.0×9.9)	3.2	6.51	1.62	285.3	1.44	1.20
Dispersion (grid)	36.0 u ² (6×6)	2.0	13.96	0.69	111.5	3.10	1.76
Dispersion (spiral)	28.3 u ² (r = 3)	2.0	17.78	0.61	111.5	3.94	1.99

Reading the table

Densification starts sub-critical. At degree 2.46 it spawns below the threshold and contracts through it at area ratio 0.55. A target of 0.40 lands at degree 6.1 and 0.73 of the critical area — a solidly connected blob, comfortably past the transition.

Dispersion at KNN 1.80 sits essentially on the fragmentation point. With the default grid spawn filling the 6×6 area, $A_c/A = 111.5 / 36.0 = 3.10$, and $\sqrt{3.10} = 1.76$. The configured target of 1.80 therefore lands the swarm just past the percolation threshold. If the spawn type is switched to Spiral the footprint becomes the inscribed circle, 28.3 u², and the equivalent target rises to 1.99.

Flocking is already connected at spawn (degree 6.15), so for it a density rule would have to be an expansion test at 1.36×, not a contraction. The goal area rule is the better fit, which is how it is configured.

Random cannot satisfy a goal area rule. Its goal area (6×6 at −6.61, −2.74) covers 36 of the 197.7 u² spawn box, so about 18% of agents begin inside it, and with cohesion 0 there is no attractor to bring in the rest. Fixed duration is correct for it.

Goal area geometry

Behaviour	Spawn centre	Goal area	Distance to travel
Flocking	(−19.30, 0.22)	6 × 26.05 at (25.29, −0.28)	41.6 u to the near edge, 44.6 u to the centre
Densification	(−0.16, −0.03)	6 × 6 at (−6.61, −2.74)	13.9 u for the farthest agent to get within 3 u of centre
Dispersion	(−6.61, −2.74)	6 × 6 at (−6.61, −2.74)	spawn and goal coincide, so a goal area rule would fire at t = 0
Random	(−0.21, 0.12)	6 × 6 at (−6.61, −2.74)	no attractor; about 18% start inside

The Flocking goal area sits beyond the right wall (Wall4 at x = 22.23), which is why Wall3 and Wall4 are in that type's walls-to-disable list. The CommonFate attractor at x = 50 provides the rightward pull.

7. Friction, top speed, and whether the targets are reachable

Friction is rule 5 of the model, an acceleration opposing the current velocity:

$$a_{\text{friction}} = -v \times f$$

Under a steady driving force F the velocity stops growing when the two balance, $F = f \times v$, giving a terminal speed of F/f . The agent is then also subject to the max speed clamp, so the speed an agent can actually sustain is:

$$\text{effective top speed} = \min(\text{maxSpeed}, F / f)$$

Raising maxSpeed above F/f does nothing

At $f = 0.9$ the friction ceiling is often the binding constraint rather than maxSpeed. For Flocking the steady driving force is the common fate attraction, whose magnitude is the cohesion intensity $c = 2$, so the ceiling is $2 / 0.9 = 2.22 \text{ u/s}$ no matter how high maxSpeed is set. To go faster than that you must raise cohesion or lower friction.

With friction $f = 0.9$, $N = 40$ and the current presets:

Behaviour	Driving force F	F / f	maxSpeed	Effective	Limited by
Flocking	2.00 (common fate, $c = 2$)	2.22	1.50	1.50	maxSpeed
Densification	5.00 (common fate, $c = 5$)	5.56	1.50	1.50	maxSpeed
Dispersion	~1.50 (separation, partly cancelling)	1.67	1.50	~1.50	marginal
Random	7.65 ($r = 20$, mean $ \text{rand} = 0.383$)	8.50	1.50	1.50	maxSpeed

So at the current maxSpeed of 1.5 friction is *not* the limiting factor for any behaviour: every F/f ceiling sits above 1.5. Dispersion is the one to watch, because its separation force is an average of unit vectors that partially cancel as the cluster becomes symmetric, so its effective F falls as it expands.

Time to reach the target

Dividing the distance each behaviour must cover by its effective speed gives a lower bound on how long the clip needs. Compared against the 20 s recordingTimePerSim:

Behaviour	Must cover	At 1.50 u/s	Verdict
Flocking	41.6 u to the goal edge	27.7 s	Exceeds the 20 s cap. Will always time out.
Flocking	44.6 u to the goal centre	29.7 s	Exceeds the cap.
Densification	13.9 u for the farthest agent	9.3 s	Comfortable.
Dispersion	2.96 u outward ($r \text{ } 3.0 \rightarrow 5.96$)	2.0 s	Comfortable, 5.3 s even if friction halves the speed.

Action needed for Flocking

The traverse from the spawn strip at $x = -19.3$ to the goal area at $x = 25.29$ is 41.6 u at minimum. At 1.5 u/s that is 27.7 s against a 20 s cap, so the clip ends on timeout with the swarm still mid-arena. Options, in order of preference:

- Set the per-type overrideRecordingTime for Flocking to 35–40 s. It already exists for exactly this case.
- Raise maxSpeed, but note the friction ceiling: at $f = 0.9$ and $c = 2$ nothing above 2.22 u/s is reachable, giving 18.7 s to the near edge and 20.1 s to the centre. That is too marginal to rely on.
- Raise Flocking cohesion, which lifts both the driving force and the ceiling: $c = 4$ gives a ceiling of 4.44 u/s and a 9.4 s traverse.
- Move the goal area closer to the spawn strip.

The 2 s warm-up delay does not help here: it is un-recorded, so the agents cover about 3 u before capture starts, leaving 41.6 u still to travel inside the recording window.

8. Where 4.51 comes from

Gilbert (1961) posed the disc model: scatter points on an infinite plane at density λ , join any two closer than R , and ask when one infinite connected component appears. This is exactly the perception graph. Because λ and R only ever enter as the product $\lambda\pi R^2$, the transition occurs at a single value of the mean degree, whatever λ and R are individually.

The constant is measured numerically, not derived. The standard high precision result is for fully penetrable discs, quoted as a critical area fraction:

$$\phi_c = 0.676339 \pm 0.000004 \text{ (Quintanilla, Torquato \& Ziff, 2000)}$$

Converting to a mean degree takes two steps. First, coverage relates to density by $\phi = 1 - \exp(-\lambda\pi r^2)$, so:

$$\lambda\pi r^2 = -\ln(1 - 0.676339) = 1.12806$$

Second, two discs of radius r overlap when their centres are within $2r$, so the connection radius is $R = 2r$ and the degree uses that radius:

$$k = \lambda\pi R^2 = \lambda\pi(2r)^2 = 4 \times \lambda\pi r^2$$

$$k = 4 \times 1.12806 = 4.512$$

Equivalently the critical density is $\lambda_c = 4.512 / \pi \approx 1.4363$ points per unit area at $R = 1$. Independent rigorous Monte-Carlo integration brackets the critical degree between 4.508 and 4.515 at 99.99% confidence, and earlier simulation work narrowed it to 4.51218–4.51228.

Assumptions worth remembering

- The constant assumes an **infinite plane and randomly scattered points**. With 40 agents in a bounded arena the transition is blurred, not sharp, so treat 4.51 as “around here cohesion is lost” rather than a hard line.
- It assumes **symmetric, omnidirectional perception**, which matches the paper's Table 1 (blind spot 0°). A non-zero blind angle makes the graph directed and the constant no longer applies directly.
- SwarmAgent measures perception to the collider **edge**, not centre to centre, which slightly inflates the effective R .
- The 0.729 constant likewise assumes a random scatter. A structured flock departs from it. Use these figures to choose between 0.4 and 4.0, not between 0.40 and 0.45.

9. Choosing a threshold in practice

1. Compute A_c for the behaviour, using the agent count and the perception radius in force.
2. Decide the endpoint in units of A_c . Below 1 is connected, above 1 is fragmenting. Roughly 0.7 of A_c gives an unambiguously cohesive blob.
3. Convert to a ratio against the spawn footprint: target ratio = target area / spawn area.
4. If using the KNN metric instead, take the square root of that ratio.
5. Press Play Motion and confirm against the live readout before committing the number.

A caution on normalising by R

Perception radius is one of the swept parameters, and participants cannot see it. If the threshold is expressed relative to R , clips recorded at different R end at different absolute densities, making endpoint appearance a systematic correlate of an invisible manipulation. Prefer using this maths to *choose* the number, then holding it fixed in absolute terms across a condition set.

10. Where this lives in the code

File	Responsibility
SwarmDensityMetrics.cs	Static maths. Monotone chain convex hull ($O(n \log n)$), shoelace area, the trim, mean KNN distance, and KNN pair collection for the gizmo.
SwarmDensityMonitor.cs	Samples the chosen metric on an interval, holds baseline / current / ratio, exposes mean degree and critical area, throttled logging, gizmos and the readout strings.
DensityAreaVisualizer.cs	Runtime LineRenderer outline of the measured hull plus the baseline hull, toggled from the panel like the perception radius.

File	Responsibility
SwarmManager.cs	Calls Evaluate each frame after the agents move; owns the Show Density Area flag.
SimRecorder.cs	DensityRatioReached end condition per motion type; captures the baseline after the warm-up delay and writes the outcome into batch_config.json.
UI.cs	Metric selector, target ratio slider, live readout and connectivity line in the play mode panel.

Recorded per clip in batch_config.json

densityMetric, densityBaseline, densityAtEnd, densityRatioAtEnd, densityTargetRatio — alongside endReason, which reads density-ratio when this rule fired and timeout when the clip ran to its cap.

11. References

- Hénard, A., Peillard, É., Rivière, J., Kubicki, S., & Coppin, G. (2024). *Human perception of swarm fragmentation*. HRI '24, Boulder, CO.
- Gilbert, E. N. (1961). *Random plane networks*. Journal of the Society for Industrial and Applied Mathematics, 9(4), 533–543.
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- Brown, D. S., Kerman, S. C., & Goodrich, M. A. (2014). *Human-swarm interactions based on managing attractors*. HRI '14, Bielefeld, Germany.